

2.0 Summative Assessment: Network Systems

Fill in the Blanks Question 1:

The (BGP) Border Gateway Protocol is known for its use in exchanging routing information between autonomous systems on the internet.

Fill in the Blanks Question 2:

In OSPF, routers within the same area share a (LSDB) Link-state Database to ensure consistent and accurate routing information.

Short Answer Question 1: Explain the term "default gateway" and its importance in a network.

A default gateway is the device—usually a router—that acts as the exit point from a local network to other networks, including the internet. When a device needs to send data to a destination outside its own network and doesn't know the exact path, it forwards that data to the default gateway for further routing.

The default gateway is located at the edge of the local network, where the internal Local Area Network (LAN) connects to external Wide Area Networks (WAN/Internet). It is typically the router's internal (LAN) IP address, like 192 . 168 . 1 . 1.

You can think of the default gateway like a main highway on-ramp. Inside your neighborhood (local network), you can navigate streets easily. But when you need to travel somewhere farther away, you take the on-ramp—the gateway—which connects you to the larger road system.

Its importance lies in enabling external communication. Without a default gateway, devices could only communicate within their local network and would have no way to reach outside resources such as websites, cloud services, or remote networks. In short, the default gateway is what allows a network to connect to the wider world.

Short Answer Question 2: Describe how distance-vector routing protocols determine the best path for data packets.

Distance-vector routing works like a community of neighbors—or even bees—sharing information about the best way to reach a destination. Each router only knows what its immediate neighbors tell it, and they regularly exchange simple updates about which networks they can reach and how far away those networks are.

When a router needs to send data, it listens to its neighbors and compares the distances they report—usually measured in hop count, or how many routers the data must pass through. It then selects the path with the shortest distance and forwards the data to the neighbor that leads in that direction.

Over time, as routers continue sharing updates, they build a clearer picture of the network. If a route becomes unavailable, they adjust by choosing the next best option based on the latest information they've received.

Like bees communicating where the closest and best pollen can be found, routers rely on shared, local knowledge to guide data along the most efficient path. In short, distance-vector routing determines the best path by learning from neighboring routers and choosing the shortest route, one step at a time.

Scenario-Based Question:

A company has multiple branch offices connected via routers using OSPF. One of the links between two routers frequently experiences outages, affecting the network stability. What steps can the network administrator take to mitigate the impact of these outages using OSPF features?

In a network using OSPF, frequent link outages can disrupt stability and performance. However, OSPF provides several built-in features that a network administrator can use to minimize the impact of these failures.

First, when a link goes down, OSPF automatically detects the failure and generates a Link-State Advertisement (LSA). This update is quickly shared with all routers in the network, allowing each router to update its network map and recalculate the best available path. This fast convergence helps maintain connectivity by rerouting traffic around the failed link.

1. To further reduce instability, the administrator can adjust the OSPF interface cost (the costs of all outbound interfaces along a path, selecting the lowest cumulative total as the best path) on the unreliable link. By increasing its cost, OSPF will prefer more stable paths under normal conditions, only using the unreliable link as a backup. This helps prevent frequent route changes caused by intermittent failures.
2. Another effective step is enabling equal-cost load balancing if multiple paths exist. This allows traffic to be distributed across multiple links, reducing reliance on any single connection and improving overall resilience.
3. The administrator can also tune hello and dead intervals to improve failure detection times. Faster detection allows routers to respond more quickly to outages, minimizing disruption.
4. Additionally, implementing redundant links or alternative paths ensures that when one link fails, traffic can continue flowing through another route without significant interruption.

Similar to a GPS system quickly rerouting when a road closes, OSPF: Detects the failed link, notifies all routers using LSAs, updates the network map, and recalculates the best path. By adjusting costs, enabling redundancy, and tuning OSPF settings, a network administrator can ensure the network remains stable even when links fail.

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